Present-epoch scale matching, the weak-field invariant $C_{\rm obs}$, and constraints on a cubic T^3 geometric construction

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The coincidence $2\pi a_0 \sim cH_0$ is a standard observation in the MOND literature. We define and test one hypothesis linking that coincidence to a *geometric derivation*: (i) a compact flat cubic three-torus of present-day side $L = c/H_0$ with a putative "circulation quantum" $\kappa = cL$, and (ii) identification of the isotropic projector ratio $\mathrm{Tr}(h)/\mathrm{Tr}(\gamma) = 2/3$ with the weak-field coefficient in an AQUAL-type action. The route is assessed against explicit microphysical, weak-field, and cosmic-scale criteria. Writing $\kappa = cL$ does not by itself quantize circulation; the package $L = c/H_0$, $\Gamma = cL$, and $\omega = H_0$ is inconsistent when ω is the cycle frequency $2\pi v/L$, without assigning curvature to a flat geodesic. The projector identity is correct but topology-independent and absent from the weak-field action, so it does not fix $C_{\rm obs}$. We instead derive the normalization-invariant response combination $C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$; the spherical response identifies only $C_{\rm obs}^2 a_0$ unless a_0 is independently fixed. At the cosmological scale, exact tracking $L_{\rm phys}(t) = c/H(t)$ is incompatible with fixed comoving moduli in ordinary FLRW except for a coasting expansion. The cited cubic E_1 bound, evaluated in its Planck-background benchmark, requires $L \gtrsim \mathcal{O}(6)\, c/H_0$, incompatible with that benchmark's $\alpha = 1$ scale match. Returning to the spherical weak-field baseline, imposing both $a_0 = cH_0/2\pi$ and the un-derived comparator $C_{\rm obs} = 2/3$ gives $a_{0,\rm eff} = (4/9)a_0$, corresponding to an approximately 18% reduction in asymptotic baryonic Tully–Fisher velocity relative to $C_{\rm obs} = 1$ at the same input a_0 and baryonic mass. This is a conditional consistency check, not a global T^3 prediction or a completed galaxy-data fit. The retained results are a present-epoch phenomenological encoding of the coincidence, a clean EFT invariant, and a reconstruction checklist for any future geometric claim.

Keywords: Modified Newtonian dynamics; cosmic topology; effective field theory; radial acceleration relation; compact manifolds

I. INTRODUCTION

Empirical galactic dynamics exhibit a characteristic acceleration $a_0 \sim 1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$ that organizes the radial acceleration relation (RAR) and the baryonic Tully–Fisher relation [1–4]. The numerical coincidence

$$2\pi\, a_0 \sim c\, H_0 \tag{1}$$

is well known [1, 2]. It is a legitimate research question whether a compact flat three-torus $T^3 = S^1 \times S^1 \times S^1$ [5, 6] and a weak-field effective field theory (EFT) can turn that coincidence into a derivation.

The distinction matters beyond topology bookkeeping. The normalization that enters the deep-MOND response also enters the RAR and the asymptotic baryonic Tully–Fisher relation, so a geometric coefficient must survive both action-level matching and observational normalization. Conversely, failure of the particular construction tested here would not decide the viability of MOND as a class, relativistic MOND theories, or other microscopic completions. It would show only that the empirical coincidence and the cubic topology do not supply the claimed derivation without additional physics.

For a definite consistency test, we define the following two-step hypothesis. It is an author-defined construction, not an attribution of this complete package to a particular published model:

(A) "Circulation quantization" on a domain of size $L = c/H_0$ yields $a_0 = cH_0/2\pi$;

(B) the transverse-projector ratio 2/3 yields the EFT coefficient $C_{\rm obs}$ in a deep-MOND square-root law.

The appeal of this construction is that a Hubble-scale domain appears to supply the dimensional acceleration input, while a projector count appears to supply its dimensionless response coefficient. If both steps followed from physical dynamics, the galactic normalization would be linked to global geometry rather than left as an independently matched input. The empirical coincidence and the trace identity make this a definite question to test; they do not themselves supply the missing phase dynamics or matter coupling. The consistency tests below ask whether the proposed shortcut survives those requirements before assigning it a physical prediction.

Our analysis identifies mathematical incompatibilities in both steps under the stated assumptions. This short technical note records those scoped constraints and isolates what *does* follow cleanly from a static AQUAL-type action [7]. The MOND coincidence, compact-topology bounds, projector algebra, and AQUAL framework are established ingredients. This note combines them in a

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scoped consistency and normalization audit. Its reusable output is a procedure for geometric/EFT identifications: distinguish a field convention from a matched coupling, state the independent acceleration input, and retain compact-source integrability before comparing forces with data. The coefficient identity is algebra within the stated action, not a new universal law or independently fixed MOND prediction; no priority claim is made.

a. Evaluation framework. The analysis uses three groups of criteria, previewed here so that each later calculation has an explicit standard: (M1)–(M4) require a phase-bearing order parameter, integer winding, a consistent meaning of L, and a dynamical map to a_0; (E1)–(E3) require action-level projector dynamics, the correct RAR/BTFR normalization, and solved compensated periodic sources; and (S1)–(S2) require a viable modulus history and consistency with compact-topology bounds. The detailed, forward-looking checklist appears in Sec. VIII. After specifying the geometry, the body proceeds from circulation (Sec. III) to the local weak-field action (Sec. IV), then to expansion history and topology bounds (Secs. V and VI). Section VII tests the resulting conditional normalization. Each sector introduces its requirements before assessing the proposed construction. Table I summarizes the resulting analytical dispositions.

TABLE I. Analytical disposition of claims discussed in this work.

Quantity / claim	**Disposition**	**Note**
Empirical coincidence $2\pi a_0 \sim cH_0$	Fact	Motivating observation only
$a_0 \equiv cH_0/2\pi$ at the present epoch	Phen.	Named postulate; not derived
"Circulation quantization $\Rightarrow a_0$"	Not derived	$\kappa = cL$ is not h/m quantization
Package $(L, \Gamma = cL, \omega = H_0)$ with $\omega = 2\pi v/L$	Incompatible	Cycle-clock inconsistency; no centripetal-force premise
Isotropic count $\mathrm{Tr}(h)/\mathrm{Tr}(\gamma) = 2/3$	Derived	Generic 3D identity; not special to T^3
$C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$	Derived	Field-rescaling invariant; only $C_{\rm obs}^2 a_0$ fixed by the spherical force
$C_{\rm obs} = 2/3$ from the trace ratio	Not derived	Projector absent from the action
$L_{\rm phys}(t) = c/H(t)$ with fixed comoving moduli	Incompatible	Needs evolving moduli or coasting FLRW
Cubic E_1 with $L = c/H_0$ vs. Planck topology	Incompatible	$L \gtrsim \mathcal{O}(6)\, c/H_0$ in the cited background/statistic
Simultaneous $a_0 = cH_0/2\pi$ and $C_{\rm obs} = 2/3$	Rescaled	Conditional spherical normalization: $a_{0,\rm eff} = (4/9)a_0$; no data fit
Compensated sources on compact T^3	Derived	$\int \mathrm{div} = 0$
Geometric derivation of a_0 or $C_{\rm obs}$	Open	Path in Sec. VIII

II. CUBIC THREE-TORUS GEOMETRY AND NOTATION

A flat three-torus is the quotient $\mathbb{R}^3/\Lambda$ by a lattice of translations [5]. The correct schematic is a parallelepiped with opposite faces identified (Fig. 1), not the doughnut surface of revolution in $\mathbb{R}^3$. That doughnut is an embedding of T^2.

"T^3" alone does not imply equal periods. Isotropic statements require a *cubic* (or explicitly isotropised) lattice. Rectangular and twisted flat manifolds (E_2, E_3, ...) are different geometries with different CMB constraints [6, 8].

Throughout, L means a *physical* edge length at the epoch under discussion unless "comoving" is written explicitly. When a numerical Hubble scale is needed we take $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ for illustration; none of the scoped arguments depends on that choice at the factor-of-two level.

III. $a_0 = cH_0/2\pi$ IS NOT CIRCULATION QUANTIZATION

a. Requirements (M1)–(M4). A circulation-based derivation must first specify a phase-bearing order parameter and mass parameter, obtain integer winding, fix the physical meaning of L, and derive its map to a_0. The requirements in Sec. VIII are the standard for the following test, rather than assumptions that the kinematic assignment already satisfies.

A. What quantization requires

For a neutral nonrelativistic superfluid with order parameter $\Psi = \sqrt{n_s}\, e^{i\theta}$, the velocity is $\mathbf{v} = (\hbar/m)\nabla\theta$, giving

$$\Gamma = \oint \mathbf{v}\cdot d\mathbf{l} = w\,\frac{h}{m}\,, \qquad w \in \mathbb{Z}\,, \tag{2}$$

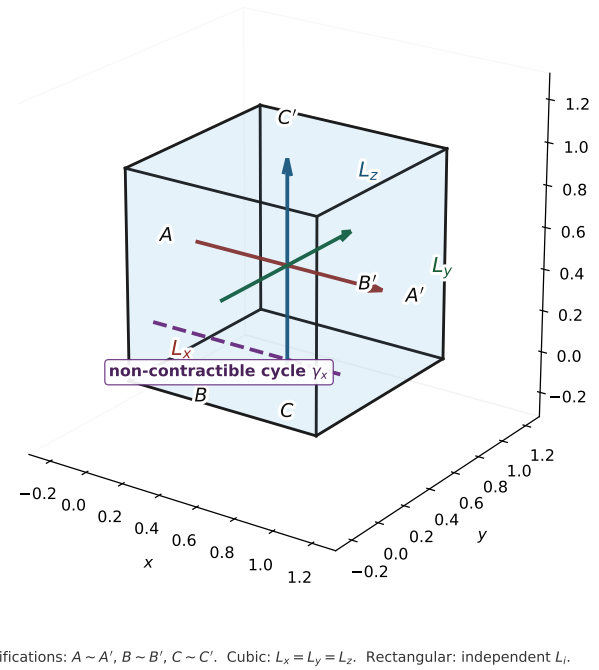


FIG. 1. Flat cubic T^3 fundamental domain: opposite faces identified ($A \sim A'$, $B \sim B'$, $C \sim C'$). Edge lengths L_x, L_y, L_z are the three independent periods. Cubic means $L_x = L_y = L_z$. This is not the embedded doughnut T^2.

with single-valuedness of Ψ on closed cycles [9–11]. Equation (2) quantizes *phase winding*, not "length times a speed limit." Relativistic circulation concerns the appropriate canonical momentum; this nonrelativistic velocity formula cannot justify a massive flow at $v = c$.

B. The author-defined circulation hypothesis

The hypothesis introduces

$$\kappa_{\rm cosmo} \stackrel{?}{=} c\,L \tag{3}$$

by multiplying a maximum speed by a length. For constant speed on a cycle of length L, one has only $\Gamma = vL$; setting $v = c$ yields $\Gamma = cL$. No integer winding, no order parameter, and no mass m enter. That is a kinematic assignment, not quantization.

If one formally equated the $w = 1$ nonrelativistic quantum to c^2/H_0,

$$m = \frac{hH_0}{c^2} \sim 10^{-32}\,\mathrm{eV}/c^2\,, \tag{4}$$

one would impose a microscopic mass relation, not eliminate microphysics. This algebraic identification is not a relativistic flow solution or a derived particle-mass prediction.

C. Internal inconsistency of the (L, Γ, ω) package

Let s denote physical arclength on a closed cycle of length $L = c/H_0$. Define a cycle clock by $\varphi = 2\pi s/L$, so $\omega = \dot\varphi = 2\pi v/L$ for uniform motion at the epoch in question. Then:

- $\Gamma = cL$ implies $v = c$;
- $\omega = H_0$ implies $v = H_0 L/(2\pi) = c/(2\pi)$, contradicting $v = c$.

Thus one cannot simultaneously maintain

$$L = \frac{c}{H_0}\,, \qquad \Gamma = cL\,, \qquad \omega = H_0 \tag{5}$$

under that cycle-clock definition. A closed geodesic of a flat torus has zero intrinsic covariant acceleration. Its periodic coordinate does not create an embedded circular orbit or a centripetal force $v^2/[L/(2\pi)]$. No such force is used in this consistency test. "Distributing cH_0 across a 2π angular topology" is not coordinate-invariant: the range of an angular chart can be rescaled. A physical 2π arises only from a defined phase convention coupled to velocity.

a. Conclusion.

$$a_0 \equiv \frac{cH_0}{2\pi} \tag{6}$$

may be retained only as a *present-epoch phenomenological scale-matching postulate* that encodes Eq. (1). For $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ one finds $a_0 \approx 1.1 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, of the observed order of magnitude [3]. Under the assumptions tested here, "geometric derivation from circulation quantization" is unsupported unless the microphysical requirements (M1)–(M4) are supplied.

IV. THE PROJECTOR RATIO DOES NOT DETERMINE $C_{\rm obs}$

a. Requirements (E1) and (E3). A geometric coefficient must be traced through an action and matter coupling to a physical force. If the projector is essential, its role and dynamics must be specified before extracting $C_{\rm obs}$, as required by (E1). Compact-source integrability and solved periodic fields remain the separate requirement (E3). The trace identity and the static action below distinguish these tasks from a count of tensor components.

A. The identity

For any n^i with $\gamma_{ij} n^i n^j = 1$, define the metric trace by $\mathrm{Tr}_\gamma(h) = \gamma^{ij} h_{ij}$. Then

$$h_{ij} = \gamma_{ij} - n_i n_j\,, \qquad \frac{\mathrm{Tr}_\gamma(h)}{\mathrm{Tr}_\gamma(\gamma)} = \frac{2}{3}\,. \tag{7}$$

This is correct, topology-independent, and independent of superfluidity. It discards tensorial orientation information and is not special to T^3.

B. Static AQUAL-type action

A static AQUAL-type force sector [7] may be written

$$S_{\rm WF} = \int dt\, d^3x \left[-\frac{|\nabla\Phi_N|^2}{8\pi G} - \rho_b \Phi_N - \frac{C_{\rm IR}}{12\pi G\, a_0} |\nabla\psi|^3 - C_m\, \rho_b\, \psi \right], \tag{8}$$

Here $a_0 > 0$, $C_m > 0$, and $C_{\rm IR} > 0$ define the attractive positive-coefficient branch used below. The potentials have dimensions $[\Phi_N] = [\psi] = \mathrm{L}^2\mathrm{T}^{-2}$, so $[|\nabla\psi|] = [a_0] = \mathrm{LT}^{-2}$, while C_m and $C_{\rm IR}$ are dimensionless. Write $q = |\nabla\psi| = \sqrt{\gamma^{ij}\partial_i\psi\partial_j\psi}$. Neither n^i nor h_{ij} appears. The kinetic operator is fully three-dimensional and isotropic. There is therefore no field-theoretic sense in which Eq. (7) "matches" onto the EFT.

Variation yields

$$\nabla\cdot\left(\frac{|\nabla\psi|}{a_0}\,\nabla\psi\right) = 4\pi G\,\frac{C_m}{C_{\rm IR}}\,\rho_b\,. \tag{9}$$

In spherical symmetry, with $g_N = |\nabla\Phi_N|$ and physical acceleration $g_P = C_m|\nabla\psi|$,

$$g_P = C_{\rm obs}\sqrt{a_0\, g_N}\,, \qquad \boxed{C_{\rm obs} = \frac{C_m^{3/2}}{\sqrt{C_{\rm IR}}}}. \tag{10}$$

C. Field-rescaling invariance

Under $\psi \mapsto s\psi$ with $s > 0$,

$$C_m \mapsto \frac{C_m}{s}\,, \qquad C_{\rm IR} \mapsto \frac{C_{\rm IR}}{s^3}\,, \tag{11}$$

while $C_{\rm obs}$ is invariant. More generally, an invariant monomial obeys $-a - 3b = 0$ for $C_m^a C_{\rm IR}^b$; the response in Eq. (10) is the positive square root of the corresponding basic ratio $C_m^3/C_{\rm IR}$. Setting $C_m = C_{\rm IR}$ so that $C_{\rm obs} = C$ is a normalization convention, not a calculation of $C = 2/3$.

Field-rescaling invariance does not make $C_{\rm obs}$ separately identifiable when the acceleration scale is also free. In the spherical force the dimensional combination is

$$a_{0,\rm eff} = \frac{C_m^3}{C_{\rm IR}} a_0 = C_{\rm obs}^2 a_0. \tag{12}$$

Indeed, $a_0 \mapsto \lambda a_0$ and $C_{\rm IR} \mapsto \lambda C_{\rm IR}$, with dimensionless $\lambda > 0$, leave Eq. (8) unchanged while $C_{\rm obs} \mapsto C_{\rm obs}/\sqrt{\lambda}$. A separate numerical $C_{\rm obs}$ therefore needs an independently fixed a_0 and matching convention; the spherical response alone cannot measure both.

a. Scope of the result. Equation (7) alone does not support a statement that $C_{\rm obs} = 2/3$ follows from T^3 geometry or from "matching." Such a derivation would require inserting h^{ij} into the action, specifying its dynamical origin, and integrating out the relevant fields.

D. Compact sources

On compact T^3, the integral of any pure divergence vanishes, so localized solutions must use a compensated density contrast (and, if present, a zero-mean exchange source). For a flat periodic metric, general morphology, and the local-static baseline with vanishing exchange source,

$$\frac{|\nabla\psi|}{a_0}\,\nabla\psi = \frac{C_m}{C_{\rm IR}}\,\nabla\Phi_N + \nabla\times\mathbf{H} + \mathbf{h}_0\,, \tag{13}$$

where $\mathbf{H}$ is periodic and $\mathbf{h}_0$ is the constant harmonic flux fixed by the solution. A periodic curl has zero spatial mean, but the nonlinear gradient flux need not, so dropping $\mathbf{h}_0$ needs an additional condition. It is not a freely fitted uniform acceleration. A nonzero exchange source requires its own source contribution before the difference is divergence-free; a curved metric requires the corresponding metric-dependent Hodge decomposition. A pointwise square-root formula is not a general identity for disks; its accuracy requires a solved field equation. Neither this decomposition nor compact topology derives $C_{\rm obs} = 2/3$.

V. PRESENT-EPOCH MATCHING VERSUS FIXED-COMOVING FLRW

a. Requirement (S1). A viable scale construction must distinguish a present-epoch identification from a dynamical length history. If the latter is intended, it must provide the corresponding modulus dynamics. The following calculation tests exact Hubble tracking with fixed comoving moduli.

For fixed comoving moduli $L^{(\rm com)}$,

$$L_{\rm phys}(t) = a(t)\, L^{(\rm com)}\,, \qquad \frac{\dot L_{\rm phys}}{L_{\rm phys}} = H\,. \tag{14}$$

Requiring instead $L_{\rm phys}(t) = c/H(t)$ gives

$$\frac{\dot L_{\rm phys}}{L_{\rm phys}} = -\frac{\dot H}{H}\,, \tag{15}$$

and equating the two yields $\dot H = -H^2$, i.e. deceleration parameter $q = 0$ (coasting). Exact tracking is therefore incompatible with ordinary matter-, radiation-, or Λ-dominated FLRW evolution under fixed comoving moduli.

The remaining alternatives are: (i) present-epoch numerical identification only; or (ii) an independent modulus sector with its own kinetic term and stress-energy. No such sector is constructed here.

The Hubble sphere c/H is also not generally a particle or event horizon; calling it a "global causal domain" therefore requires an additional causal definition [12].

VI. CUBIC TOPOLOGY BOUNDS AND $L = c/H_0$

a. Requirement (S2). The global length must also satisfy topology constraints evaluated for the same declared background and statistic. This test is separate from the modulus-history requirement in Sec. V; the benchmark below does not turn a background-dependent bound into a universal exclusion.

Planck topology analyses constrain cubic E_1 three-tori within the background and signal models used in those analyses. In terms of the radius R_i of the largest sphere inscribed in the fundamental domain, the likelihood analysis reports $R_i > 0.97\,\chi_{\rm rec}$ at a log-likelihood-ratio threshold $\Delta \ln \mathcal{L} > -5$ relative to the simply connected flat Planck best-fit model [6]. The polarization matched-circles search gives the same numerical radius limit at 99% confidence; these are distinct statistics, not interchangeable confidence statements. For a cube of side L, $R_i = L/2$, so

$$L \gtrsim 1.94\,\chi_{\rm rec}\,. \tag{16}$$

With $\chi_{\rm rec} \simeq 3.15\,c/H_0$ in a Planck cosmology [13],

$$L \gtrsim 6\,\frac{c}{H_0} \tag{17}$$

as an approximate conversion in that background, not a topology-only inequality in Hubble units. Recent topology surveys likewise place the shortest E_1 side near the diameter scale of the last-scattering surface, not one Hubble radius [6, 8].

Write $L = \alpha\,c/H_0$. Empirical scale matching uses $\alpha = 1$; cubic bounds in the stated benchmark require $\alpha \gtrsim \mathcal{O}(6)$. If the phenomenological formula is reinterpreted as $a_0 = c^2/(2\pi L) = cH_0/(2\pi\alpha)$, an α satisfying that benchmark bound yields

$$a_0 \lesssim \frac{1}{6} \times \frac{cH_0}{2\pi} \sim 1.8 \times 10^{-11}\,\mathrm{m\,s^{-2}}\,, \tag{18}$$

which no longer reproduces the intended MOND-scale match. Therefore the cubic-E_1, $\alpha = 1$ package fails that simultaneous scale-matching and topology benchmark.

Vague statements that a "Hubble-scale domain is compatible with non-detection" are not supported for cubic E_1 under these bounds. An altered expansion history, recombination distance, or CMB transfer model requires recomputing the distance conversion and applicable likelihood; the factor of approximately six is not an all-cosmology exclusion of T^3.

VII. CONDITIONAL RAR NORMALIZATION CHECK IN THE SPHERICAL BASELINE

a. Requirement (E2). The matched response must retain the intended RAR/BTFR normalization. This final check connects the proposed geometric inputs to their consequence for the analytic force and velocity normalization.

This section is deliberately conditional. It uses the static AQUAL-type action of Eq. (8), the positive-mass spherical branch, regular-origin flux, and no added homogeneous flux. It then imposes $C_{\rm obs} = 2/3$ as a comparator; Sec. IV does not derive that value. The result is therefore neither a global periodic-T^3 solution nor a disk-galaxy fit.

In the deep-MOND regime, $g_P = C_{\rm obs}\sqrt{a_0 g_N}$ is equivalent to

$$g_P = \sqrt{a_{0,\rm eff}\,g_N}\,, \qquad a_{0,\rm eff} = C_{\rm obs}^2\,a_0\,. \tag{19}$$

Imposing both $a_0 = cH_0/2\pi \approx 1.1 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and $C_{\rm obs} = 2/3$ yields

$$a_{0,\rm eff} = \frac{4}{9}a_0 \approx 4.8 \times 10^{-11}\,\mathrm{m\,s^{-2}}\,, \tag{20}$$

whereas modern RAR determinations find $a_0 \simeq 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ [3, 4], a factor ~ 2.5 normalization difference under these assumptions. In the asymptotic deep-MOND spherical baseline, $v^4 \propto C_{\rm obs}^2 a_0 G M_b$. At fixed input a_0 and baryonic mass, the unit-response comparator gives

$$\frac{v(C_{\rm obs} = 2/3)}{v(C_{\rm obs} = 1)} = \sqrt{\frac{2}{3}} \simeq 0.8165. \tag{21}$$

The approximately 18% reduction is specifically this same-scale comparator; it does not use the empirical reference $a_{\rm emp} = 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. For that distinct reference and the illustrative $H_0 = 70$ input, the velocity ratio is $\left[a_{0,\rm eff}/a_{\rm emp}\right]^{1/4} \simeq 0.7957$, or a reduction of about 20.4%. These are analytic comparators, not observed discrepancies or completed empirical rejections.

Within this branch, one cannot simultaneously identify $a_0 = cH_0/2\pi$ with the empirical MOND scale and impose a universal $C_{\rm obs} = 2/3$ without changing the effective normalization. Consistent options for a future fit are:

1. set $C_{\rm obs} = 1$ (up to the definition of a_0);
2. redefine the input scale so that $C_{\rm obs}^2 a_0$ matches the RAR;
3. drop universality and derive a morphology-dependent response from a solved disk PDE before fitting.

This note stops at analytic normalization. It contains no data likelihood, nuisance marginalization, intrinsic-scatter test, or model evidence. Such inferences require a follow-up with a specified geometry-dependent response, matching inputs, data selection, and nuisance model.

VIII. EVALUATION CRITERIA, RETAINED RESULTS, AND A RECONSTRUCTION PATH

A. What this note establishes

1. The coincidence (1) is empirical motivation only.
2. $a_0 \equiv cH_0/2\pi$ is allowed solely as a *present-epoch phenomenological postulate*.
3. The weak-field combination $C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$ is a derived, normalization-invariant response coefficient at fixed a_0 [Eq. (10)]; without that choice the spherical force identifies only $C_{\rm obs}^2 a_0$.
4. On compact T^3, sources must be compensated and disk solutions generally admit a curl term [Eq. (13)].
5. The scoped incompatibilities in Secs. III–VII constrain the tested cubic geometric packaging of those two numbers without deciding MOND or alternative microscopic constructions as a class.

B. Path required before any geometric-derivation claim

A subsequent work that claims to *derive* a_0 or $C_{\rm obs}$ from topology or microphysics must close the following chain. Items are grouped by sector, and each is a separately testable technical requirement.

a. Microphysics and circulation.

(M1) Define a condensate order parameter $\Psi = \sqrt{\rho}\, e^{i\theta}$ (or a relativistic analogue) with a definite mass parameter m.

(M2) Obtain the circulation from $\nabla\theta$ and impose integer winding on non-contractible cycles of T^3, yielding $\Gamma = w\, h/m$ in the nonrelativistic limit rather than $\Gamma = cL$.

(M3) Fix a single geometric meaning of L (cube edge, closed geodesic length, or circumference) and use it consistently in every formula.

(M4) Derive a map $(\Gamma, m, w, L) \mapsto a_0$ that does not rely on a chart-dependent "distribution" of cH_0 over an angular range of 2π.

b. Cosmic scale and moduli.

(S1) Either restrict scale matching to the present epoch, or supply a modulus action for $L_{\rm phys}(t)$ that does not force the coasting condition $q = 0$.

(S2) Confront the resulting L with Planck/COMPACT bounds for cubic E_1 and for twisted flat alternatives [6, 8].

c. Weak-field coefficient and data.

(E1) If a projector h_{ij} is essential, insert it into the action, specify the dynamics of n^i, and compute $C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$ as a definite number independent of field normalization.

(E2) Reconcile $C_{\rm obs}^2 a_0$ with the empirical RAR/BTFR normalization, accounting for the conditional factor 4/9 in Sec. VII rather than treating it as an independent data rejection.

(E3) Solve the compensated periodic nonlinear equation for at least one locally spherical and one disk baryonic source, including curl and harmonic contributions; periodicity does not supply global spherical symmetry.

d. Representation (already satisfied here).

(R1) Use a face-identified parallelepiped for T^3 (Fig. 1), never an embedded doughnut T^2.

Absent evidence closing (M1)–(M4), (S1)–(S2), and (E1)–(E3), this note supports present-epoch phenomenological matching of Eq. (1), the derived invariant $C_{\rm obs}$, and the scoped constraints on the tested cubic construction—not a geometric derivation of the MOND scale.

CODE AND DATA AVAILABILITY

Figure 1 is produced by `Scripts/itsm_t3_fundamental_domain.py`. Source code is available in the public GitHub repository and the project Zenodo archive. Scripts that render an embedded doughnut surface depict T^2 and should not be used as T^3 schematics. No new observational catalogue is analysed.

DECLARATION ON THE USE OF ARTIFICIAL-INTELLIGENCE TOOLS

Artificial-intelligence-assisted tools were used for editorial drafting, source-code assistance, and checks of symbolic and numerical consistency. This local review draft still requires the author's final verification of the manuscript, equations, cited sources, and generated artifacts. Submission requires the author's confirmation of that review and acceptance of full responsibility; neither is certified by these automated checks.

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